

OPENHMI REFERENCE MODULE

TCW40

Allwinner A40i

English Technical Reference Manual

A practical choice for mature embedded HMI products that need broad display and industrial I/O support without moving to a higher-cost application processor.

Revision 1.1 | August 2026

Supplier-neutral reference compiled from the supplied module documentation. Confirm the exact hardware revision, pin mapping, BSP status and commercial configuration before design-in.

1. Positioning and selection

This manual summarizes the module from a project-selection perspective. It is intended to help product teams decide whether the platform fits their HMI architecture before requesting detailed carrier-board, software and supply-chain information.

When this module is a good fit

- Cost-sensitive industrial panels and instrumentation
- Products with established Linux or Android 7.1 software
- Designs requiring parallel RGB, LVDS, MIPI DSI or HDMI output
- Projects that value broad legacy I/O and dual-Ethernet options

When to consider another path

- New products requiring long-term support for a current Android release
- High-resolution 3D user interfaces or compute-intensive vision workloads
- Designs that require native PCIe or USB 3.0 expansion

Selection rule

Choose the module only after checking display interface, graphics load, boot-time target, operating-system baseline, carrier-board complexity, production lifecycle and the engineering capability available to your team.

2. Core specifications

Processor	Allwinner A40i; quad-core Arm Cortex-A7
Graphics	Mali-400 MP2; OpenGL ES 2.0 and OpenVG 1.1
Memory	DDR3, typically 1 GB; up to 2 GB by configuration
Storage	eMMC, typically 8 GB; documented support up to 64 GB
Operating systems	Linux and Android 7.1 support are documented
Module interface	180 castellated edge pins
Input power	4.8 V to 5.5 V DC; 5 V nominal; USB VBUS supply option

Configuration note

Memory, storage, wireless and some peripheral options are configuration-dependent. The values above describe documented capability, not a guaranteed stock configuration. Confirm the required combination before quotation or schematic release.

Software baseline

Operating-system support depends on the available BSP, kernel branch, display driver, touch controller, GPU stack and peripheral drivers. Validate the actual software package on target hardware before committing the product schedule.

3. Display, video and touch

Panel outputs	LVDS, parallel RGB/CPU and 4-lane MIPI DSI; functions are multiplexed
External video	HDMI 1.4, CVBS and VGA/YPbPr options
Resolution	Up to 1920 x 1080 is documented for LVDS/RGB and HDMI-class output
Multi-display	Two independent output paths are documented for selected interface combinations
Camera/video input	Parallel camera, BT.656, CVBS and YPbPr options; CSI up to 720p30
Touch	I2C interface suitable for common capacitive-touch controllers

Display integration checks

- Confirm that the required panel timing, lane count and voltage level are supported by the selected pin-multiplexing mode.
- Check whether dual-display use reduces access to camera, Ethernet, storage or general-purpose I/O functions.
- Validate backlight power, enable and PWM control on the carrier board; these are not supplied by the display data interface alone.
- Confirm the touch-controller interface, reset/interrupt pins, firmware and display bonding stack for the intended panel.
- Run the real UI workload on the target resolution to verify frame rate, memory bandwidth and thermal behavior.

4. Connectivity and I/O

USB	1 x USB 2.0 OTG and 2 x USB 2.0 host
Ethernet	One 10/100/1000 MAC and one 10/100 MAC; dual-port operation is documented
Serial buses	Up to 7 UART, 4 SPI and 4 TWI/I2C through multiplexed pins
Control	Up to 6 PWM outputs, ADC key inputs, infrared controller and RTC support
Audio	Microphone, line input, stereo output, I2S and S/PDIF options
Wireless	Optional Wi-Fi or combined Wi-Fi/Bluetooth configuration
Other	SD/MMC and optional lithium-battery input/charge monitoring

Pin-multiplexing warning

Interface counts describe the documented processor and module capability. Several functions share pins, clocks or high-speed resources and therefore cannot always be used together. Build a project-specific I/O allocation table before starting the carrier-board schematic.

Request before layout

Obtain the current pin table, carrier-board reference schematic, PCB footprint, power sequence and BSP release note for the exact module revision selected for the project.

5. Mechanical, power and design-in

PCB thickness	1.6 mm
Top-side component height	Approximately 1.8 mm above the module PCB
Carrier keep-out	A carrier-board cutout is required under the bottom-side component area
Footprint	Confirm the released mechanical drawing and module revision before carrier layout

Power and handling

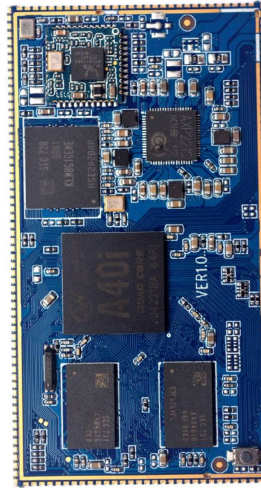
- Do not hot-plug the module or connect/disconnect it while the carrier board is powered.
- Use a regulated supply with startup margin for display, USB, Ethernet, wireless and storage peak current.
- Follow the documented power sequence, reset timing and signal-voltage domains.
- Provide airflow or heat spreading after measuring the real application workload and enclosure temperature.
- Protect the assembly from condensation, conductive contamination, ESD and mechanical stress.

Project validation checklist

- Exact module revision and production configuration confirmed
- Display and touch panel tested with the intended cable and bonding stack
- All pin conflicts resolved in a carrier-board I/O matrix
- BSP, boot media, update method and recovery path verified
- Thermal, EMC, ESD, power interruption and long-duration operation tested
- Lifecycle, lead time and alternative-component strategy reviewed

6. Hardware views and reference drawings

The photographs below are reproduced from the supplied module documentation. Supplier marks have been redacted while component placement, connectors and edge contacts are left unchanged. Photographs are for identification only and must not be scaled for PCB or enclosure design.



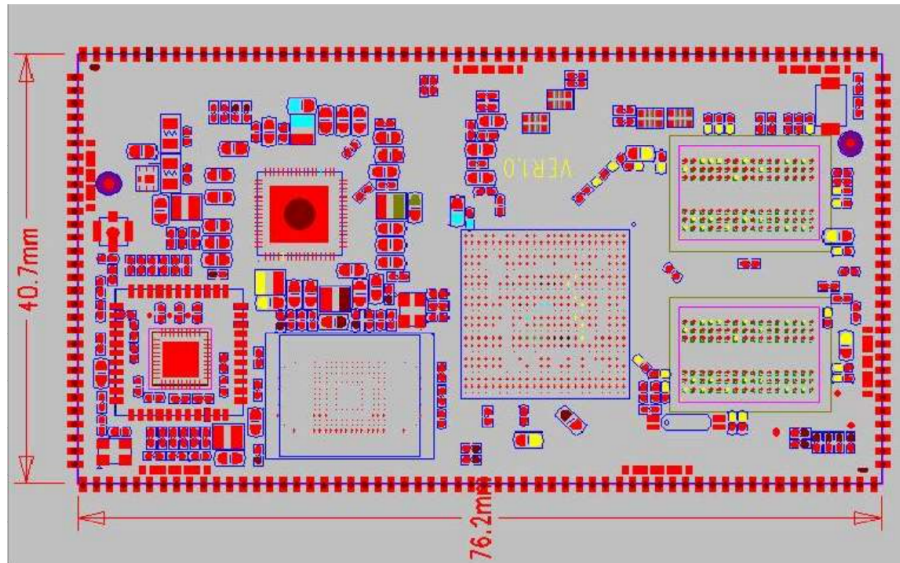
Module top view

Mechanical drawing

5. 物理尺寸图

5.1 核芯板正面高度 1.8MM,底部限高 0MM 中间元件部位需掏空底板, 板厚 1.6MM。

顶部



底部

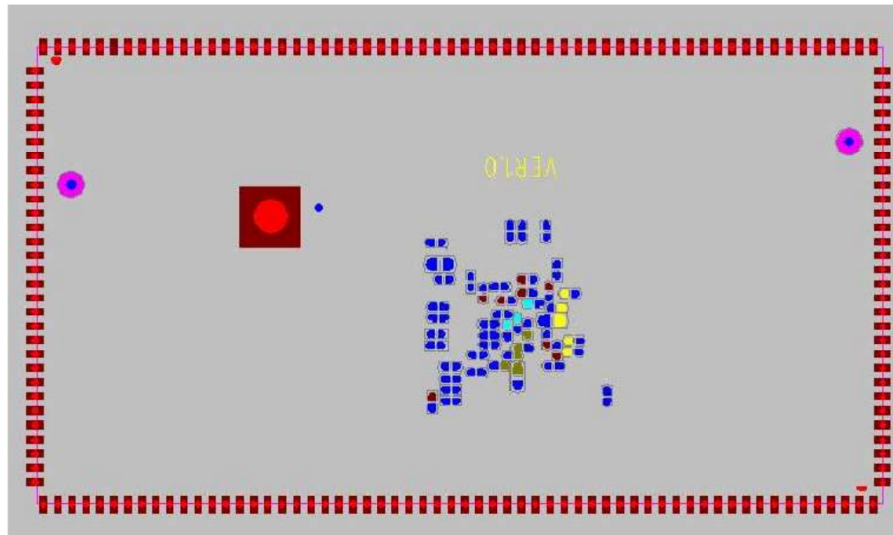


Figure: Mechanical drawing

7. Pin definitions and multiplexing reference

The following pages reproduce the original engineering pin tables at high resolution. Signal identifiers and alternate-function names are preserved exactly to reduce transcription risk. Use them as an evaluation reference; the controlled pin table for the selected hardware revision remains the design authority.

English legend for recurring source labels

PIN / GPIO	Physical module pin and associated GPIO bank or signal
Function 1...n	Selectable alternate function through SoC pin multiplexing
Input / Output / I/O	Signal direction: input, output, or bidirectional
Power input / Power output	Supply rail entering or leaving the module
GND	Power and signal ground
NC / Floating	Do not connect unless a controlled revision document states otherwise
Highlighted cells	Default or reserved module functions in the source table; confirm before reassignment

Important

Never route a carrier board from a web manual alone. Request the revision-matched pin spreadsheet, schematic, footprint and power-domain table before schematic sign-off.

Pin definition reference - sheet 1 of 4

A6	PF5	SDC0_D2	JTAG_CK1			
A7	PI15/EINT27	SDC0-DET	SPI1_CS1	PS2_SDA1	TCLKIN1	
A8	PG0	CSII_PCLK	TS1_CLK	SDC1_CMD		
A9	PG1	CSII_MLCK	TS1_ERR	SDC1_CLK		
A10	PG3	CSII_VSYNC	TS1_DVLD	SDC1_D1		
A11	PG2	CSII_HSYNC	TS1_SYNC	SDC1_D0		
A12	PG4	CSII_D0	TS1_D0	SDC1_D2	CSI0_D8	
A13	PG5	CSII_D1	TS1_D1	SDC1_D3	CSI0_D9	
A14	PG6	CSII_D2	TS1_D2	UART3_TX	CSI0_D10	
A15	PG7	CSII_D3	TS1_D3	UART3_RX	CSI0_D11	
A16	PG8	CSII_D4	TS1_D4	UART3_RTS	CSI0_D12	
A17	PG9	CSII_D5	TS1_D5	UART3_CTS	CSI0_D13	BIST_RESULT0
A18	PG10	CSII_D6	TS1_D6	UART4_TX	CSI0_D14	BIST_RESULT1
A19	PG11	CSII_D7	TS1_D7	UART4_RX	CSI0_D15	
A20	PI2	TWI4_SCK				
A21	PI3	TWI4_SDA	PWM1			
A22	PI0	TWI3_SCK				
A23	PI1	TWI3_SDA				
A24	PI12/EINT24	32KFOUT	SPI0_MOSI	UART6_TX	CLK_OUT_A	
A25	PI10/EINT22	UART5_TX	SPI0_CS0			
A26	PI11/EINT23	UART5_RX	SPI0_CLK			
A27	PI13/EINT25	USB0-DRVVBUS	SPI0_MISO	UART6_RX	CLK_OUT_B	
A28	PI14/EINT26	USB0-VBUSDET	SPI0_CS1	PS2_SCK1	TCLKIN0	
A29	PI18/EINT30	UART2_TX	SPI1_MOSI			
A30	PI19/EINT31	UART2_RX	SPI1_MISO			
A31	PI20	CTP-RST	PS2_SCK0	UART7_TX	PWM2	
A32	PI17/EINT29	CTP-INT	SPI1_CLK	UART2_CTS		
A33	PI21	LCB-BL-EN	PS2_SDA0	UART7_RX	PWM3	
A34	PA16	ECOL	CAN_TX	UART1_DCD	GCLKIN/ECOL	I2S1_DO
A35	PA17	ETXERR	CAN_RX	UART1_RING	GNUL/ETXERR	I2S1_DI
A36	PA15	ECRS	UART7_RX	UART1_DSR	GTXC/ECRS	I2S1_LRCK
A37	PA13	ETXEN	UART6_RX	UART1_CTS	GTXC/ETXEN	
A38	PA14	ETXCK	UART7_TX	UART1_DTR	GNUL/ETXCK	I2S1_BCLK
A39	PA12	EMDIO	UART6_TX	UART1_RTS	GMDIO	
A40	PA11	EMDC		UART1_RX	GMDC	
A41	PA10	ERXDV		UART1_TX	GRXC/ERXDV	
A42	PA9	ERXERR	SPI3_CS1		GNUL/ERXERR	I2S1_MCLK
A43	PA8	ERXCK	SPI3_MISO		GRXCK	
A44	PA7	ETXD0	SPI3_MOSI		GTXD0	
A45	PA6	ETXD1	SPI3_CLK		GTXD1	
A46	PA5	ETXD2	SPI3_CS0		GTXD2	
A47	PA4	ETXD3	SPI1_CS1		GTXD3	
A48	PA3	ERXD0	SPI1_MISO	UART2_RX	GRXD0	
A49	PA2	ERXD1	SPI1_MOSI	UART2_TX	GRXD1	
A50	PA1	ERXD2	SPI1_CLK	UART2_CTS	GRXD2	
A51	PA0	ERXD3	SPI1_CS0	UART2_RTS	GRXD3	

Original engineering table, source page 7. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 2 of 4

A52	PB14	SPI2_CS0	JTAG_MS0			
A53	PB15	SPI2_CLK	JTAG_CK0			
A54	PB16	SPI2_MOSI	JTAG_DO0			
A55	PB17	SPI2_MISO	JTAG_DI0			
A56	PB18	CTP-SCK	TWII_SCK			
A57	PB19	CTP-SDA	TWII_SDA			
A58	PB20	TWI2_SCK		PWM4		
A59	PB21	TWI2_SDA		PWM5		
A60	PB2	LCD-PWM	PWM0			

B 端口如下:

PIN	GPIO 功能	功能 1	功能 2	功能 3	功能 4	功能 5
B1	GND					
B2	GND					
B3	+5V 输入					
B4	+5V 输入					
B5	USBVBUS(+5V)输入					
B6	+3.3V 输出					
B7	VCC-LCD(3.3V)输出给屏用					
B8	USB-DP2					
B9	USB-DM2					
B10	USB-DP1					
B11	USB-DM1					
B12	USB-DP0					
B13	USB-DM0					
B14	PHONEOUTP	POWER-KEY				
B15	PHONEOUTN	RESET				
B16	VMIC					
B17	MICIN2					
B18	MICIN1					
B19	LINEINL					
B20	LINEINR					
B21	HPOUTL					
B22	HPOUTR					
B23	KEYADC0					
B24	HAGND					
B25	TVIN0					
B26	TVIN1					
B27	TVIN2					
B28	TVIN3					
B29	VGA-G	TVOUT0				
B30	VGA-B	TVOUT1				
B31	VGA-R	TVOUT2				
B32	GND					
B33	PD24	LCD0_CLK	SMC_VCCEN			
B34	PD25	LCD0_DE	SMC_RST			

Original engineering table, source page 8. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 3 of 4

B35	PD26	LCD0_HSYNC	SMC_SLK			
B36	PD27	LCD0_VSYNC	SMC_SDA			
B37	PD23	LCD0_D23(R7)	SMC_DET			
B38	PD22	LCD0_D22(R6)	SMC_VPPPP			
B39	PD21	LCD0_D21(R5)	SMC_VPPEN			
B40	PD20	LCD0_D20(R4)	CSI1_MCLK			
B41	PD19	LCD0_D19(R3)	LVDS1_VN3	MIPI-DSI-D3N		
B42	PD18	LCD0_D18(R2)	LVDS1_VP3	MIPI-DSI-D3P		
B43	PD17	LCD0_D17(R1)	LVDS1_VNC	MIPI-DSI-D2N		
B44	PD16	LCD0_D16(R0)	LVDS1_VPC	MIPI-DSI-D2P		
B45	PD15	LCD0_D15(G7)	LVDS1_VN2	MIPI-DSI-D1N		
B46	PD14	LCD0_D14(G6)	LVDS1_VP2	MIPI-DSI-D1P		
B47	PD13	LCD0_D13(G5)	LVDS1_VN1	MIPI-DSI-D0N		
B48	PD12	LCD0_D12(G4)	LVDS1_VP1	MIPI-DSI-D0P		
B49	PD11	LCD0_D11(G3)	LVDS1_VN0	MIPI-DSI-CKN		
B50	PD10	LCD0_D10(G2)	LVDS1_VP0	MIPI-DSI-CKP		
B51	PD9	LCD0_D9(G1)	LVDS0_VN3			
B52	PD8	LCD0_D8(G0)	LVDS0_VP3			
B53	PD7	LCD0_D7(B7)	LVDS0_VNC			
B54	PD6	LCD0_D6(B6)	LVDS0_VPC			
B55	PD5	LCD0_D5(B5)	LVDS0_VN2			
B56	PD4	LCD0_D4(B4)	LVDS0_VP2			
B57	PD3	LCD0_D3(B3)	LVDS0_VN1			
B58	PD2	LCD0_D2(B2)	LVDS0_VP1			
B59	PD1	LCD0_D1(B1)	LVDS0_VN0			
B60	PD0	LCD0_D0(B0)	LVDS0_VP0			

C 端口如下:

PIN	GPIO 功能	功能 1	功能 2	功能 3	功能 4	功能 5
C1	PI4	SDC3_CMD				
C2	PI5	SDC3_CLK				
C3	PI6	SDC3_D0				
C4	PI7	SDC3_D1				
C5	PI8	SDC3_D2				
C6	PI9	SDC3_D3				
C7	PI16/EINT28	SPI1_CS0	UART2_RTS	SDC3-DET		
C8	PB3	PWM1	SPDIF_MCLK			
C9	PB4	IR0_RX				
C10	PB9	I2S_DO1		PWM6		
C11	PB10	I2S_DO2		PWM7		
C12	PB11	I2S_DO3				
C13	PB22	UART0_TX				
C14	PB23	UART0_RX	IR1_RX			
C15	PH4/EINT4	LCD1_D4		UART4_TX		
C16	PH5/EINT5	LCD1_D5		UART4_RX		
C17	PH6/EINT6	LCD1_D6		UART5_TX		
C18	PH7/EINT7	LCD1_D7		UART5_RX		

Original engineering table, source page 9. Signal names are preserved; see the English legend on the preceding page.

Pin definition reference - sheet 4 of 4

C19	PH8/EINT8	LCD1_D8	ERXD3	KP_IN0		
C20	PH9/EINT9	LCD1_D9	ERXD2	KP_IN1		
C21	PH11/EINT11	LCD1_D11	ERXD0	KP_IN3		
C22	PH12/EINT12	LCD1_D12		PS2_SCK1		
C23	PH13/EINT13	LCD1_D13		PS2_SDA1	SMC_RST	
C24	PH14/EINT14	LCD1_D14	ETXD3	KP_IN4	SMC_VPPEN	
C25	PH19/EINT19	LCD1_D19	ERXERR	KP_OUT1	SMC_SDA	
C26	PH20/EINT20	LCD1_D20	ERXDV	CAN_TX		
C27	PH21/EINT21	LCD1_D21	EMDC	CAN_RX		
C28	ADC1	FEL(BOOT)				
C29	CVBS	TVOUT3				
C30	AVCC(3V)输出					

D 端口如下:

PIN	GPIO 功能	功能 1	功能 2	功能 3	功能 4	功能 5
D1	PB5	I2S_MCLK	AC97_MCLK			
D2	PB8	I2S_DO0	AC97_DO		PCM-DOOUT	
D3	PB6	I2S_BCLK	AC97_BCLK		PCM-CLK	
D4	PB12	I2S_DI	AC97_DI	SPDIF_DI	PCM-DIN	
D5	PB7	I2S_LRCK	AC97_SYNC		PCM-SYNC	
D6	PB13	SPI2_CS1	SPDIF_DO			
D7	PH1/EINT1	LCD1_D1		UART3_RX		
D8	PH3/EINT3	LCD1_D3		UART3_CTS		
D9	PH2/EINT2	LCD1_D2		UART3_RTS		
D10	PH0/EINT0	LCD1_D0		UART3_TX		
D11	GND					
D12	WIFI/BT_ANT					
D13	GND					
D14	HECE					
D15	HHPD					
D16	HSCL					
D17	HSDA					
D18	HTX2P					
D19	HTX2N					
D20	HTX1P					
D21	HTX1N					
D22	HTX0P					
D23	HTX0N					
D24	HTXCP					
D25	HTXCN					
D26	GND					
D27	PS_OUT(可以做外设供电用, 电源等于输入电压, 带过流保护和电量检测)。					
D28	PS_OUT(可以做外设供电用, 电源等于输入电压, 带过流保护和电量检测)。					
D29	VBAT(3.4V-4.2V)锂电池电压					
D30	VBAT(3.4V-4.2V)锂电池电压					

Original engineering table, source page 10. Signal names are preserved; see the English legend on the preceding page.

8. Document scope and next step

This is an English selection and integration reference. It intentionally excludes supplier branding, contact details, service claims and unverified commercial statements. Original photos and engineering-table facsimiles are included where they help identification or pin evaluation, with supplier marks redacted. This manual is not a replacement for the controlled pinout, schematic, mechanical drawing, processor datasheet or BSP documentation for a released production configuration.

Recommended next step

Share the product application, display size and interface, target resolution, operating system, required I/O, annual volume range, environmental conditions and development capability. The module can then be matched against the real project requirements and compared with chip-level or more integrated HMI paths.

Discuss a similar project

Visit openhmi.network to request relevant platform, display, touch and engineering expertise for your application.

Document control

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